SIMPLEX METHOD: BIG M AND TWO PHASE

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Minimize $z = 4x_1 + x_2$

Subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Introduce two artificial slack variables R_1 and R_2 and consider a very big number M.

$$\text{Minimize } z = 4x_1 + x_2 + MR_1 + MR_2$$

Subject to

$$3x_1 + x_2 + R_1 = 3$$

$$4x_1 + 3x_2 - x_3 + R_2 = 6$$

$$x_1 + 2x_2 + x_4 = 4$$

$$x_1, x_2, x_3, x_4, R_1, R_2 \geq 0$$

As M is very large, solution of it is likely to make R_1 and R_2 equal to 0.

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z	−4	−1	0	−M	−M	0	0
R_1	3	1	0	1	0	0	3
R_2	4	3	−1	0	1	0	6
x_4	1	2	0	0	0	1	4

Make the table consistent by multiplying second and third rows by *M* and adding in z-row.

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z	−4+7M	−1+4M	–M	0	0	0	9M
R_1	3	1	0	1	0	0	3
R_2	4	3	−1	0	1	0	6
x_4	1	2	0	0	0	1	4

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z	0	(1+5M)/3	-M	(4-7M)/3	0	0	4+2M
x_1	1	1/3	0	1/3	0	0	1
R_2	0	5/3	−1	-4/3	1	0	2
x_4	0	5/3	0	-1/3	0	1	3

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z	0	0	1/5	(8-5M)/5	(-1-5M)/5	0	18/5
x_1	1	0	1/5	3/5	-1/5	0	3/5
x_2	0	1	-3/5	-4/5	3/5	0	6/5
x_4	0	0	1	1	-1	1	1

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z	0	0	0	(39-25M)/5	-M	-1/5	17/5
x_1	1	0	0	2/5	0	-1/5	2/5
x_2	0	1	0	-1/5	0	3/5	9/5
x_3	0	0	1	1	-1	1	1

$$x_1 = 2/5, x_2 = 9/5, x_3 = 1$$

$$z = 17/5$$

Two Phase method of Simplex

First phase:

Minimize $r=R_1+R_2$

$$\text{Subject to}$$

$$3x_1 + x_2 + R_1 = 3$$

$$4x_1 + 3x_2 - x_3 + R_2 = 6$$

$$x_1 + 2x_2 + x_4 = 4$$

$$x_1, x_2, x_3, x_4, R_1, R_2 \geq 0$$

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
r	0	0	0	−1	−1	0	0
R_1	3	1	0	1	0	0	3
R_2	4	3	−1	0	1	0	6
x_4	1	2	0	0	0	1	4

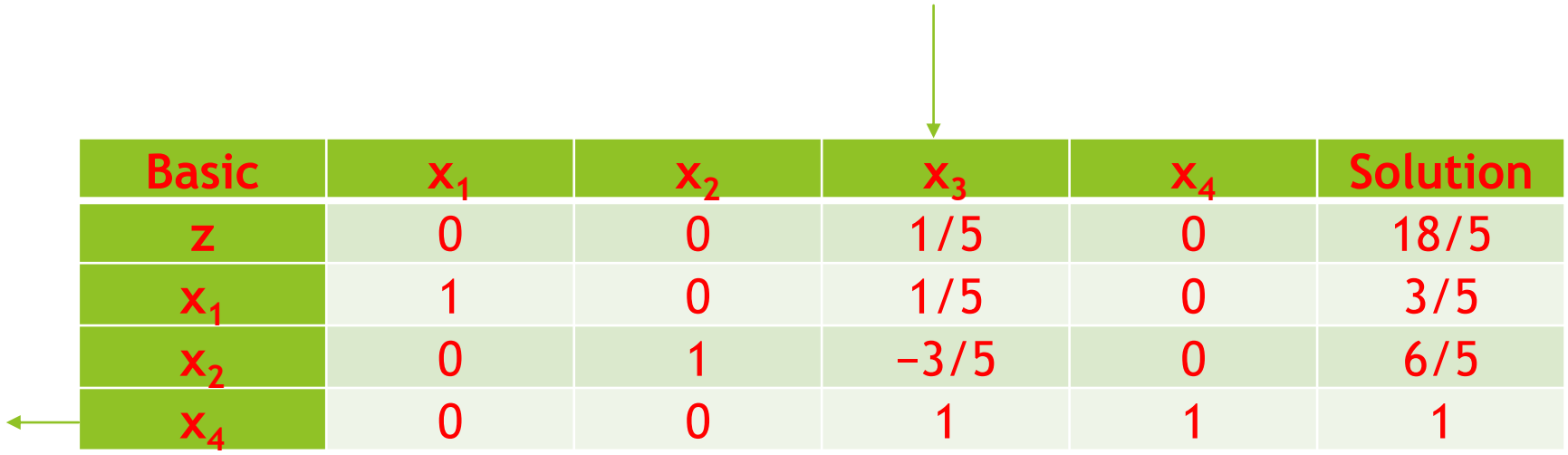
Make the *r*-row consistent by adding second and third row in *r*-row. Apply the simplex table, you shall get the following final table:

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
r	0	0	0	−1	−1	0	0
x_1	1	0	1/5	3/5	−1/5	0	3/5
x_2	0	1	−3/5	−4/5	3/5	0	6/5
x_4	0	0	1	1	−1	1	1

$$x_1 = 3/5, x_2 = 6/5, x_4 = 1$$

Basic	x_1	x_2	x_3	x_4	Solution
z	-4	-1	0	0	0
x_1	1	0	1/5	0	3/5
x_2	0	1	−3/5	0	6/5
x_4	0	0	1	1	1

Make it consistent.



Basic	x_1	x_2	x_3	x_4	Solution
z	0	0	1/5	0	18/5
x_1	1	0	1/5	0	3/5
x_2	0	1	−3/5	0	6/5
x_4	0	0	1	1	1

Basic	x_1	x_2	x_3	x_4	Solution
z	0	0	0	0	17/5
x_1	1	0	0	-1/5	2/5
x_2	0	1	0	3/5	9/5
x_3	0	0	1	1	1

Solve the following problem using two-phase Simplex method and big M method:

Maximize $Z=15A+20B$

Subject to

$A+2B \geq 10$

$2A-3B \leq 6$

$A+B \geq 6$

$A \geq 0$ and $B \geq 0$

Two-phase Simplex method

Phase I: Minimizing the Artificial Variable

Define the objective function: $r = R_1 + R_2$

subject to

$A + 2B - C + R_1 = 10$

$2A - 3B + D = 6$

$A + B - E + R_2 = 6$

$A, B, C, D, E, R_1, R_2 \geq 0$

Basic	A	B	C	D	E	R_1	R_2	Solution
r	0	0	0	0	0	−1	−1	0
R_1	1	2	−1	0	0	1	0	10
D	2	−3	0	1	0	0	0	6
R_2	1	1	0	0	−1	0	1	6

Basis	A	B	C	D	E	R_1	R_2	Solution
r	2	3	−1	0	−1	0	0	16
R_1	1	2	−1	0	0	1	0	10
D	2	−3	0	1	0	0	0	6
R_2	1	1	0	0	−1	0	1	6

Basis	A	B	C	D	E	R_1	R_2	Solution
r	2	3	−1	0	−1	0	0	16
B	1/2	1	−1/2	0	0	1/2	0	5
D	7/2	0	−3/2	1	0	3/2	0	21
R_2	1/2	0	1/2	0	−1	−1/2	1	1

Basis	A	B	C	D	E	R_1	R_2	Solution
r	1/2	0	1/2	0	−1	−3/2	0	1
B	0	1	−1	0	1	1	−1	4
D	0	0	−5	1	7	5	−7	14
R_2	1	0	1	0	−2	−1	2	2

Basis	A	B	C	D	E	R_1	R_2	Solution
r	0	0	0	0	0	−1	−1	0
B	0	1	−1	0	1	1	−1	4
D	0	0	−5	1	7	5	−7	14
A	1	0	1	0	−2	−1	2	2

Phase 2

Basis	A	B	C	D	E	Solution
Z	-15	-20	0	0	0	0
B	0	1	−1	0	1	4
D	0	0	−5	1	7	14
A	1	0	1	0	−2	2

Basis	A	B	C	D	E	Solution
Z	0	0	−5	0	−10	110
B	0	1	−1	0	1	4
D	0	0	−5	1	7	14
A	1	0	1	0	−2	2

Basis	A	B	C	D	E	Solution
Z	0	0	−5	0	−10	110
B	0	1	−1	0	1	4
D	0	0	−5	1	7	14
A	1	0	1	0	−2	2

Basis	A	B	C	D	E	Solution
Z	0	0	−5	0	−10	110
B	0	1	−2/7	−1/7	0	2
E	0	0	−5/7	1/7	1	2
A	1	0	−3/7	2/7	0	6

Basis	A	B	C	D	E	Solution
Z	0	0	−85/7	10/7	0	130
B	0	1	−2/7	−1/7	0	2
E	0	0	−5/7	1/7	1	2
A	1	0	−3/7	2/7	0	6

Based on the optimality condition, one of the coefficients in the Z-row is negative. Since C has a negative coefficient (−85/7), it suggests that entering C into the basis could improve the solution. However, the issue is that no variable can leave the basis, meaning the solution is unbounded, i.e., the objective function can increase indefinitely without reaching an optimal solution.